

Report
Of
NetSage AI: AI-Assisted Cisco Network Troubleshooting Engine with
Deterministic Validation and Human-in-the-Loop Review

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410031414

Name of Student: VIBHU MAURYA

Batch: 2026-27

CANDIDATE'S DECLARATION

I, VIBHU MAURYA, do hereby declare that the internship report titled "NetSage AI: AI-Assisted Cisco Network Troubleshooting Engine with Deterministic Validation and Human-in-the-Loop Review" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: VIBHU MAURYA

Roll No.: 2410031414

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous mentorship and technical guidance throughout the design, development, and evaluation of this project.

I express my gratitude to my parents for their blessings.

Signature

Student Name: VIBHU MAURYA

Roll No.: 2410031414

INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



एन सी ई आर
एन सी ई आर
एन सी ई आर



NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

VIBHU MAURYA

IILM University, Greater Noida

has successfully completed the 8-weeks

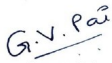
Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS



Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 4f8aadd0e6a55ff94cbf
Student ID: STU676c4543504781735148867



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-59 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

TABLE OF CONTENTS

Chapter No.	Chapter Name	Page No.
	Cover Page	---
1.	Candidate's Declaration	
2.	Acknowledgment	
3.	Internship Completion Certificate	
4.	Project Description	
	4.1 Introduction	
	4.2 Organization Profile	
	4.3 Problem Statement	
	4.4 Project Objectives	
	4.5 Scope of the Project	
	4.6 Technologies and Tools Used	
	4.7 System Architecture	
	4.8 Methodology	
	4.9 Expected Outcomes	
	4.10 Certificates of Completion and Communication Proof	
5.	Bibliography/References	

4. Project Description

4.1 Introduction

Modern enterprise networks are expected to deliver near-zero downtime and high operational resilience. In practice, however, diagnosing multi-layer network anomalies — spanning VLAN and trunk misconfigurations, dynamic routing neighbour failures, DHCP pool exhaustion, ACL shadowing, and NAT address pool depletion — remains a cognitively demanding and time-intensive task, particularly for junior network engineers, NOC technicians, and Tier-1/Tier-2 support staff.

This report documents "NetSage AI: AI-Assisted Cisco Network Troubleshooting Engine with Deterministic Validation and Human-in-the-Loop Review", a project undertaken as part of the internship programme. The project, named NetSage AI, is a hybrid networking-intelligence system designed to bridge the gap between automated fault detection and human operational governance. It combines a deterministic rule-based verification engine, a generative-AI reasoning pipeline built on Google Gemini 2.5 Flash, an interactive chatbot and diagnostic-studio backend, and a mandatory Human-in-the-Loop (HITL) review protocol supported by an analytics dashboard.

NetSage AI was evaluated against 35 representative enterprise network fault scenarios, simulated end-to-end in Cisco Packet Tracer, and covering all major layers of the OSI model. The goal of the project was to demonstrate that pairing fast, verifiable rule-based checks with generative reasoning — under continuous human review — can deliver rapid, explainable, and auditable root-cause analysis, diagnostic next-step recommendations, and exact Cisco IOS remediation commands.

4.2 Organization Profile

This project was undertaken as part of a Cisco-affiliated internship programme, carried out through the Cisco Networking Academy in association with the School of Computer Science and Engineering, IILM University, Greater Noida. Cisco Networking Academy is Cisco Systems' global IT and networking skills-development programme, which provides curriculum, certification pathways, and simulation tools such as Cisco Packet Tracer to learners worldwide, with the objective of building practical, industry-relevant networking and IT competencies.

Within this programme, the simulated enterprise network used throughout the project was built entirely in Cisco Packet Tracer, modelling a realistic campus-branch topology comprising edge and branch routers, access/distribution switches, a wireless access point, and core network services (DHCP and DNS), all configured using genuine Cisco IOS commands and conventions. The internship work was carried out under faculty

mentorship at IILM University, from problem framing through design, implementation, and evaluation, while Google's Gemini 2.5 Flash generative-AI model, accessed via the google-genai SDK, served as the additional external AI reasoning engine underpinning the diagnostic pipeline.

4.3 Problem Statement

Junior network engineers often know individual Cisco IOS commands but struggle to connect an observed symptom to its real root cause. When, for example, a PC obtains an IP address but still cannot reach a server, the underlying fault could lie in VLAN assignment, routing, DHCP, DNS, ACL policy, or NAT — and distinguishing between these possibilities from raw show-command output is slow and highly dependent on individual expertise.

The assignment brief for this project ("NetSage AI — Build an AI troubleshooting helper with human review", under the Modern AI course, Networking Labs domain) required building a troubleshooting assistant for Cisco-style lab networks that uses symptoms, Packet Tracer topology notes, and show-command outputs to recommend a likely fault, the affected OSI layer, the next diagnostic command to run, and an evidence-backed fix — with the explicit safety rule that a human reviewer must approve or correct every AI diagnosis before it is accepted. Relying purely on an AI model to diagnose and remediate faults without this safeguard is risky, since large language models can produce confident but incorrect (hallucinated) root causes, especially when evidence is incomplete or a plausible-but-wrong explanation exists.

4.4 Project Objectives

In line with the assignment brief, the project set out to build and demonstrate the following components:

- Case dataset: assemble at least 30 troubleshooting cases drawn from Packet Tracer/lab scenarios, covering VLAN, gateway, DHCP, DNS, routing, ACL, NAT, and wireless issues — each with its symptom, topology note, show-command outputs, expected fault, OSI layer, and concept tag.
- AI prompt library: design structured prompts that force JSON output with fields for `root_cause`, `confidence`, `evidence`, `next_command`, and `fix_steps`, grounded with worked few-shot examples.
- Rule checker: implement a Python script with deterministic checks for common configuration mistakes (duplicate IPs, wrong subnet masks, gateway mismatches, interfaces down, missing VLANs, missing routes, and more), run before/alongside the AI diagnosis.

- Human review protocol: mark every AI diagnosis as Accepted, Edited, or Rejected, and maintain a Responsible-AI log documenting at least 5 cases where the AI's answer had to be corrected by a human reviewer, with the reasoning explained.
- Dashboard: build a simple, visual summary of issue types, severity, and the AI-versus-human agreement rate.
- End-to-end demo: show one broken lab case being diagnosed by the AI, reviewed by a human, fixed, and verified using the appropriate Cisco IOS commands.

NetSage AI fulfilled and extended these objectives, ultimately covering 35 benchmark cases (exceeding the minimum of 30) with a 20-rule deterministic checker, a Gemini 2.5 Flash-based AI prompt pipeline, a full Human-in-the-Loop review workflow, a Responsible-AI audit log of 5 corrected cases, and an interactive analytics dashboard with a live diagnostic studio and chatbot copilot, going beyond a static dashboard to a real-time REST API-driven system.

4.5 Scope of the Project

The scope of the project covers the design, implementation, and evaluation of the NetSage AI diagnostic pipeline within a simulated network environment. It includes:

- A simulated campus-branch enterprise topology built in Cisco Packet Tracer, consisting of two edge/branch routers, four switches, one wireless access point, and DHCP/DNS services across five VLANs.
- 35 benchmark fault cases covering VLAN and trunking issues, inter-VLAN and dynamic/static routing, physical and interface faults, DHCP services, DNS resolution, ACL/security policy, NAT/PAT, IP addressing, wireless (WLAN) configuration, and configuration persistence.
- A 20-rule deterministic checker, a Gemini-based AI diagnosis engine, a human review and audit workflow, a chatbot/REST API server, and an analytics dashboard.

The project is scoped to a simulated Cisco IOS environment. Direct integration with live production devices over SSH/NETCONF, multi-vendor support, and automated closed-loop remediation (applying fixes without human approval) are outside the current scope and are identified as future enhancements.

4.6 Technologies and Tools Used

Category	Technology / Tool
Network simulation	Cisco Packet Tracer

Programming language	Python 3
Generative AI model	Google Gemini 2.5 Flash
AI SDK	google-genai (native v1beta client)
Backend / API server	Custom Python multi-threaded HTTP server (chatbot_server.py)
Frontend / Dashboard	HTML5, CSS, JavaScript, Chart.js
Data storage / interchange	CSV (case bank, results), JSON (per-case AI responses)
Configuration management	.env-based API key management
Diagnostic evidence sources	Cisco IOS show/debug command outputs

4.7 System Architecture

NetSage AI is built as a five-tier modular pipeline designed for high accuracy, zero unvetted automated configuration changes, strict output determinism, and full auditability. The five tiers are:

- Simulated Network Testbed — the Cisco Packet Tracer topology and the 35 benchmark fault cases, from which CLI show-command outputs and topology notes are extracted.
- Hybrid Diagnostic Pipeline — CLI outputs are passed both to the deterministic rule checker (rule_checker.py) and to the AI diagnosis engine (run_diagnosis.py).
- Governance and Responsible-AI Audit — rule violations and raw AI predictions are validated by a human reviewer (add_human_review.py) and compiled into an audited results file and a responsible-AI audit log.
- Real-Time API and Chatbot Server — a local REST API and chatbot server (chatbot_server.py, chatbot.py) exposes the AI diagnosis engine for live, conversational troubleshooting.
- Interactive Analytics and Diagnostic Studio — a dashboard (dashboard.html) presents KPIs, case-level results, and a live diagnostic studio with an embedded AI copilot.

The simulated network topology used to generate the benchmark cases consists of an edge router (Router 1) performing NAT/PAT, OSPF routing, ACL filtering, and DHCP service; a branch router (Router 2) serving remote subnets; a core switch (Switch 1) trunked to access switches (SW2, SW3, SW4) and a wireless access point; and five VLANs — Engineering (VLAN 10), Management (VLAN 20), Finance (VLAN 30), Marketing (VLAN 40), and an isolated Guest Wi-Fi segment (VLAN 50).

4.8 Methodology

The project followed a five-stage methodology, matching the pipeline architecture described above:

- Simulation and data extraction — 35 network faults were recreated in Cisco Packet Tracer. Device show-command outputs (show ip interface brief, show vlan brief, show ip route, show access-lists, show ip nat translations, show ip dhcp pool, and others) and observed symptoms were structured into a benchmark case file.
- Deterministic pre-filtering — a 20-rule Python engine performed regex-based pattern matching across the CLI outputs, catching clear-cut configuration errors (for example, administratively shut down interfaces, missing dot1Q encapsulation on subinterfaces, or incorrect access VLANs) in under 0.05 seconds across the entire benchmark.
- Generative AI reasoning — cases not fully resolved by the deterministic rules, or involving multi-layer interactions, were passed to Gemini 2.5 Flash with a structured JSON-schema prompt and few-shot examples, returning a root cause, confidence level, cited evidence, next diagnostic command, and exact remediation commands.
- Human-in-the-loop review — a human reviewer compared every AI diagnosis against the known ground-truth fault for each case and labelled it Accepted, Edited, or Rejected, recording the reasoning for any correction in a dedicated audit log.
- Interactive operation and analytics — validated results were compiled into a results file and rendered on an interactive dashboard; engineers could additionally query the system live through the chatbot copilot or the REST API.

The system exposes four REST endpoints on a local server: `/api/cases` (returns the benchmark case bank), `/api/chat` (multi-turn conversational assistance), `/api/diagnose` (structured real-time diagnosis), and `/api/rules/check` (runs the deterministic rule engine). The deployment workflow consists of setting the Gemini API key in an environment file, running the rule checker and AI diagnosis scripts across all cases, merging human review annotations, generating the dashboard, and finally starting the chatbot server for live use.

A key part of the methodology was the Responsible-AI audit. Of the 35 automated diagnoses, human review accepted 25 (71.4%) as fully accurate, edited 5 (14.3%) where the AI identified the correct area but missed an important nuance, and rejected 5 (14.3%) where the AI proposed a speculative or incorrect cause. Representative edited/rejected cases included an interface reported as simply "administratively shut down" when the actual state was BPDU-Guard err-disabled; an ACL flagged as the

cause of dropped pings when the real cause was a downstream routing loop; NAT inside/outside roles flagged as inverted when the NAT pool was in fact exhausted; a generic Layer-1 fault reported when the actual cause was a duplex mismatch; and a "configuration not saved" diagnosis where the real cause was a configuration register set to bypass NVRAM (0x2142). These cases were used to refine the evidence requirements in the AI prompt and to reinforce the necessity of mandatory human review.

4.9 Expected Outcomes

The completed system demonstrated the following measurable outcomes across the 35-case benchmark:

Metric	Result
Total benchmark cases	35
AI diagnosis acceptance rate	71.4% (25 / 35)
Human-edited rate	14.3% (5 / 35)
AI rejection rate	14.3% (5 / 35)
Deterministic rules implemented	20
Average rule-engine execution time	< 1.5 ms per case
AI confidence distribution	91.4% High, 8.6% Medium
OSI layer coverage	Layers 1 through 7

By OSI layer, the benchmark was weighted toward Layer 3 issues (16 cases, 45.7%, mainly routing, subinterface encapsulation, DHCP and NAT/PAT), followed by Layer 2 (10 cases, 28.6%, VLAN/trunking and wireless mapping), Layer 4 (6 cases, 17.1%, ACL/port filtering), and Layer 7 (3 cases, 8.6%, DNS). By severity, 3 cases (8.6%) were rated Critical, 24 (68.6%) High, 6 (17.1%) Medium, and 2 (5.7%) Low.

The project confirmed several practical engineering lessons: deterministic pre-filtering is essential for catching obvious misconfigurations instantly and cheaply before invoking the AI model; prompt specificity and few-shot grounding in the literal show-command evidence substantially reduce speculative AI answers; human verification remains non-negotiable, since AI models can overlook physical-layer counters or low-level register states that a rule or a human would catch; and every proposed remediation should include a verification step (for example, re-running show ip interface brief or show ip route) to confirm the fix actually resolved the fault.

Overall, NetSage AI validates the practical viability of a hybrid AI-plus-rule-based architecture for network diagnosis. Future enhancements identified for the system include direct SSH/NETCONF telemetry integration (using tools such as Netmiko or

Scrapli) for live device polling, automated validation of proposed fixes inside sandboxed Packet Tracer instances before deployment, and Retrieval-Augmented Generation (RAG) over enterprise runbooks and historical incident tickets to further improve AI diagnostic accuracy.

4.10 Certificates of Completion and Other Communication Mail Proof

The following certificates were earned in support of this internship and project, evidencing the underlying skill development in AI and data-driven problem solving:

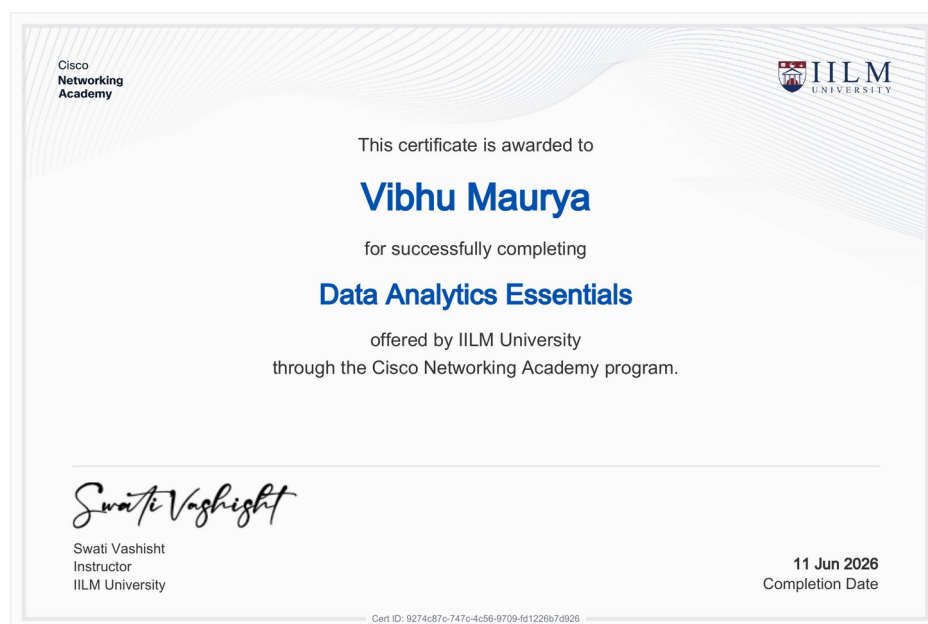
Introduction to Modern AI — Cisco Networking Academy, IILM University (26 May 2026)



Apply AI: Analyze Customer Reviews — Cisco Networking Academy, IILM University (03 June 2026)



Data Analytics Essentials — Cisco Networking Academy, IILM University (11 June 2026)



5. Bibliography/References

1. Cisco Systems Inc., "Cisco Packet Tracer Documentation," Cisco Networking Academy.
2. Google, "Gemini API Documentation," Google AI for Developers, ai.google.dev.
3. Python Software Foundation, "Python 3 Documentation," docs.python.org.
4. Chart.js Contributors, "Chart.js Documentation," chartjs.org.
5. Cisco Systems Inc., Cisco IOS Command Reference (show and debug commands).
6. Cisco Networking Academy, CCNA Curriculum Materials.